

BEC303 Network Analysis — Module 4 Quiz

Laplace Transformation and Applications

Field	Value
Subject code	BEC303
Module	4 — Laplace Transformation and Applications
Duration	45 minutes
Maximum marks	50 (Part A 20 + Part B 30)
Attempts allowed	1
Passing marks	20
Textbook	W. H. Hayt, J. E. Kemmerly and S. M. Durbin, Engineering Circuit Analysis, 8th ed., 2012

Instructions

- Part A carries 20 one-mark multiple choice questions. Exactly one option is correct.
- Part B carries 13 short questions totalling 30 marks.
- Transform tables are not supplied; standard transforms are expected to be known.
- State the region of convergence or the pole locations wherever they affect validity.
- Non-programmable scientific calculators are permitted.

Course outcome and Bloom coverage

Band	Questions	Share	Target
L1–L2 Remember / Understand	7	21%	~30%
L3–L4 Apply / Analyse	17	52%	~45%
L5–L6 Evaluate / Create	9	27%	~25%

Course outcome spread: CO4: 33. Total questions: 33.

Part A — Multiple choice (20 × 1 mark = 20 marks)

A1. The Laplace transform of the unit step $u(t)$ is [C04 | L1 | 1 mark | easy | Ch.14, pp.543-546]

- (A) $1/s$
- (B) 1
- (C) s
- (D) $1/s^2$

A2. The Laplace transform of $e^{-at} \cdot u(t)$ is [C04 | L1 | 1 mark | easy | Ch.14, pp.543-546]

- (A) $1/(s + a)$
- (B) $1/(s - a)$
- (C) $a/(s^2 + a^2)$

- (D) $s/(s + a)$

A3. The differentiation theorem states that df/dt transforms to [C04 | L2 | 1 mark | medium | Ch.14, pp.553-561]

- (A) $sF(s) - f(0-)$
- (B) $sF(s)$
- (C) $F(s)/s$
- (D) $sF(s) + f(0-)$

A4. In the s -domain, a capacitor of value C has impedance [C04 | L1 | 1 mark | easy | Ch.15, pp.571-578]

- (A) $1/(sC)$
- (B) sC
- (C) C/s
- (D) s/C

A5. An inductor with initial current $i(0-)$ is modelled in the s -domain as sL in series with [C04 | L3 | 1 mark | medium | Ch.15, pp.571-578]

- (A) a voltage source of value $L \cdot i(0-)$
- (B) a voltage source of value $i(0-)/s$
- (C) a current source of value $L \cdot i(0-)$
- (D) a short circuit

A6. The final-value theorem may be applied only if [C04 | L4 | 1 mark | hard | Ch.14, pp.561-564]

- (A) all poles of $sF(s)$ lie strictly in the left half plane
- (B) $F(s)$ is a proper rational function
- (C) the numerator degree exceeds the denominator degree
- (D) the circuit contains no capacitors

A7. For $F(s) = (3s + 6)/(s(s + 3))$, the initial value $f(0+)$ is [C04 | L3 | 1 mark | medium | Ch.14, pp.561-564]

- (A) 3
- (B) 2
- (C) 6
- (D) 0

A8. Partial fraction expansion of $(s + 4)/(s(s + 2))$ gives [C04 | L3 | 1 mark | medium | Ch.14, pp.546-553]

- (A) $2/s - 1/(s + 2)$
- (B) $1/s + 1/(s + 2)$
- (C) $2/s + 1/(s + 2)$
- (D) $4/s - 2/(s + 2)$

A9. The transfer function $H(s)$ is defined with [C04 | L3 | 1 mark | medium | Ch.15, pp.588-589]

- (A) all initial conditions set to zero
- (B) initial conditions included as sources
- (C) the load disconnected
- (D) s set equal to $j\omega$

A10. The poles of a network function are [C04 | L2 | 1 mark | medium | Ch.15, pp.588-589]

- (A) its natural frequencies
- (B) the frequencies it blocks completely
- (C) always real and negative
- (D) determined by the input waveform

A11. The step response transform of a system with transfer function $H(s)$ is [C04 | L2 | 1 mark | easy | Ch.14, pp.543-546]

- (A) $H(s)/s$
- (B) $H(s)$
- (C) $H(s)/s^2$
- (D) $s \cdot H(s)$

A12. A triangular pulse rising from 0 to 1 over $0 \leq t \leq 1$ and falling to 0 by $t = 2$ is synthesised as [C04 | L4 | 1 mark | hard | TB2 §6.8]

- (A) $t \cdot u(t) - 2(t-1)u(t-1) + (t-2)u(t-2)$
- (B) $t \cdot u(t) - (t-1)u(t-1)$
- (C) $t \cdot u(t) - (t-1)u(t-1) + (t-2)u(t-2)$
- (D) $u(t) - 2u(t-1) + u(t-2)$

A13. Convolution in the time domain corresponds in the s -domain to [C04 | L2 | 1 mark | medium | Ch.15, pp.589-598]

- (A) multiplication
- (B) addition
- (C) convolution again
- (D) division

A14. For an RC low-pass with $R = 1 \text{ k}\Omega$ and $C = 1 \text{ }\mu\text{F}$, the pole is at [C04 | L3 | 1 mark | medium | Ch.15, pp.598-602]

- (A) $s = -1000 \text{ rad/s}$
- (B) $s = -159.2 \text{ rad/s}$
- (C) $s = -0.001 \text{ rad/s}$
- (D) $s = +1000 \text{ rad/s}$

A15. The -3 dB cutoff frequency of that same RC low-pass, in hertz, is [C04 | L3 | 1 mark | medium | Ch.15, pp.598-602]

- (A) 159.2 Hz
- (B) 1000 Hz
- (C) 6283 Hz
- (D) 0.159 Hz

A16. For a first-order filter the product of bandwidth in hertz and 10–90 % rise time in seconds is approximately [C04 | L5 | 1 mark | hard | Ch.15, pp.598-602]

- (A) 0.35
- (B) 1.0
- (C) 2.2
- (D) 6.28

A17. Among poles at $s = -2000$ and $s = -100 \pm j400$, the dominant pole is [C04 | L5 | 1 mark | hard | Ch.15, pp.602-606]

- (A) the complex pair at $-100 \pm j400$
- (B) the real pole at -2000
- (C) whichever has the largest magnitude
- (D) they contribute equally

A18. A pole located exactly on the $j\omega$ axis corresponds to [C04 | L3 | 1 mark | medium | Ch.15, pp.602-606]

- (A) an undamped sustained oscillation
- (B) an exponentially decaying response
- (C) an exponentially growing response
- (D) no response at all

A19. Cascading two identical unbuffered RC low-pass sections gives a transfer function that is [C04 | L5 | 1 mark | hard | Ch.15, pp.606-610]

- (A) not the square of the single-section $H(s)$, because of loading
- (B) exactly the square of the single-section $H(s)$
- (C) the sum of the two section transfer functions
- (D) independent of the second section

A20. A passive driving-point impedance must be a positive real function, which requires [C04 | L5 | 1 mark | hard | Ch.15, pp.606-610]

- (A) no right-half-plane poles or zeros and $\text{Re}\{Z\} \geq 0$ for $\text{Re}\{s\} \geq 0$
- (B) all poles on the $j\omega$ axis
- (C) the numerator degree to exceed the denominator degree
- (D) only real coefficients

Part B — Descriptive (13 questions, 30 marks)

B1. State the differentiation theorem and explain how it makes initial conditions automatic in circuit analysis. [C04 | L3 | 3 marks | Ch.14, pp.553-561]

B2. Transform $2 \frac{di}{dt} + 6i = 12u(t)$ with $i(0^-) = 3$ A, solve for $I(s)$, and find $i(t)$. [C04 | L4 | 3 marks | Ch.14, pp.546-561]

B3. State both value theorems, apply them to $F(s) = (3s + 6)/(s(s + 3))$, and state the condition that governs the final-value theorem. [C04 | L4 | 3 marks | Ch.14, pp.561-564]

B4. Derive the transfer function of a series RC low-pass with the output across the capacitor, and relate its pole to the time constant and the cutoff frequency. [C04 | L4 | 3 marks | Ch.15, pp.588-602]

B5. Explain how an inductor and a capacitor with non-zero initial conditions are modelled in the s -domain. [C04 | L3 | 2 marks | Ch.15, pp.571-578]

B6. Perform the partial fraction expansion of $(s + 4)/(s(s + 2))$ and write the time function. [C04 | L3 | 2 marks | Ch.14, pp.546-553]

B7. Explain the relationship between the impulse, step and ramp responses, both in time and in the s -domain. [C04 | L3 | 2 marks | Ch.14, pp.543-546]

- B8.** Synthesise the transform of a triangular pulse that rises from 0 to 1 over $0 \leq t \leq 1$ and falls back to 0 by $t = 2$. [C04 | L4 | 2 marks | TB2 §6.8]
- B9.** Convolve $h(t) = e^{-t}u(t)$ with $x(t) = u(t)$, and verify the result in the s-domain. [C04 | L5 | 2 marks | Ch.15, pp.589-598]
- B10.** Given poles at $s = -2000$ and $s = -100 \pm j400$, describe the natural response and identify the dominant pole. [C04 | L5 | 2 marks | Ch.15, pp.602-606]
- B11.** Explain why the bandwidth–rise time product is constant for a first-order circuit, and state its value. [C04 | L5 | 2 marks | Ch.15, pp.598-602]
- B12.** Realise $H(s) = 1000/(s + 1000)$ as a passive RC network and explain why cascading two such sections does not give $H(s)^2$. [C04 | L5 | 2 marks | Ch.15, pp.606-610]
- B13.** Explain why the s-domain approach is preferred over the time-domain approach for a driven second-order circuit with non-zero initial conditions. [C04 | L5 | 2 marks | Ch.15, pp.578-585]

Answer key — Part A

Q	Answer	CO	Bloom	Reference
A1	A	CO4	L1	Ch.14, pp.543-546
A2	A	CO4	L1	Ch.14, pp.543-546
A3	A	CO4	L2	Ch.14, pp.553-561
A4	A	CO4	L1	Ch.15, pp.571-578
A5	A	CO4	L3	Ch.15, pp.571-578
A6	A	CO4	L4	Ch.14, pp.561-564
A7	A	CO4	L3	Ch.14, pp.561-564
A8	A	CO4	L3	Ch.14, pp.546-553
A9	A	CO4	L3	Ch.15, pp.588-589
A10	A	CO4	L2	Ch.15, pp.588-589
A11	A	CO4	L2	Ch.14, pp.543-546
A12	A	CO4	L4	TB2 §6.8
A13	A	CO4	L2	Ch.15, pp.589-598
A14	A	CO4	L3	Ch.15, pp.598-602
A15	A	CO4	L3	Ch.15, pp.598-602
A16	A	CO4	L5	Ch.15, pp.598-602
A17	A	CO4	L5	Ch.15, pp.602-606
A18	A	CO4	L3	Ch.15, pp.602-606
A19	A	CO4	L5	Ch.15, pp.606-610
A20	A	CO4	L5	Ch.15, pp.606-610

Part A — why each option is right or wrong

A1. Correct answer: (A) $1/s$

- (A) Correct — integrating the kernel from 0 to infinity gives $1/s$.
- (B) Wrong - 1 is the transform of the unit impulse.
- (C) Wrong - s corresponds to differentiation, not to a step.
- (D) Wrong - $1/s$ squared is the transform of the ramp.

A2. Correct answer: (A) $1/(s + a)$

- (A) Correct — with a pole at $s = -a$ matching the decay rate.
- (B) Wrong - that pole is in the right half plane and corresponds to a growing exponential.
- (C) Wrong - that is the transform of a sinusoid, not an exponential.
- (D) Wrong - this is an improper form whose inverse would include an impulse.

A3. Correct answer: (A) $sF(s) - f(0-)$

- (A) Correct — the $f(0-)$ term is exactly how the initial condition enters the algebra.
- (B) Wrong - dropping the initial-condition term forces a separate and error-prone boundary step later.
- (C) Wrong - dividing by s is the integration theorem.
- (D) Wrong - the sign is inverted, which flips the direction of the initial transient.

A4. Correct answer: (A) $1/(sC)$

- (A) Correct — the capacitor law integrates, so its impedance divides by s .
- (B) Wrong - sC makes the capacitor behave like an inductor and inverts the frequency response.
- (C) Wrong - the capacitance belongs in the denominator with s , not the numerator.
- (D) Wrong - this is dimensionally incorrect.

A5. Correct answer: (A) a voltage source of value $L i(0-)$

- (A) Correct — this is the source form of the $f(0-)$ term from the differentiation theorem.
- (B) Wrong - that is the capacitor form, where the initial voltage divides by s .
- (C) Wrong - the series model requires a voltage source; the current-source form goes in parallel and has value $i(0-)/s$.
- (D) Wrong - a short discards the stored energy entirely.

A6. Correct answer: (A) all poles of $sF(s)$ lie strictly in the left half plane

- (A) Correct — a pole on the $j\omega$ axis or in the right half plane means no final value exists, though the limit still returns a number.
- (B) Wrong - properness matters for the inverse transform but does not govern the final-value theorem.
- (C) Wrong - that condition would make the function improper and is not the validity test.
- (D) Wrong - capacitors are irrelevant; what matters is where the poles lie.

A7. Correct answer: (A) 3

- (A) Correct — $sF(s) = (3s+6)/(s+3)$, which tends to 3 as $s \rightarrow \infty$.
- (B) Wrong - 2 is the final value, obtained as s tends to zero.
- (C) Wrong - 6 is the numerator constant and is not a limit of the expression.
- (D) Wrong - zero would require the numerator degree to be lower than it is.

A8. Correct answer: (A) $2/s - 1/(s + 2)$

- (A) Correct — cover-up at $s = 0$ gives 2 and at $s = -2$ gives -1 .

- (B) Wrong - both residues are wrong; the cover-up evaluations were not performed.
- (C) Wrong - the sign of the second residue is inverted.
- (D) Wrong - these residues do not reconstruct the original numerator.

A9. Correct answer: (A) all initial conditions set to zero

- (A) Correct — $H(s)$ is the zero-state ratio and is a property of the network alone.
- (B) Wrong - including them makes the ratio depend on stored energy, so it is no longer a network property.
- (C) Wrong - the load is part of the network whose transfer function is being defined.
- (D) Wrong - substituting $j\omega$ gives the frequency response, a special case of $H(s)$.

A10. Correct answer: (A) its natural frequencies

- (A) Correct — they are the roots of the characteristic equation and set the natural response.
- (B) Wrong - blocked frequencies are the zeros, the roots of the numerator.
- (C) Wrong - poles are frequently complex, which is what produces ringing.
- (D) Wrong - poles belong to the network and are independent of the excitation.

A11. Correct answer: (A) $H(s)/s$

- (A) Correct — the step transform is $1/s$, so the output is $H(s)/s$.
- (B) Wrong - $H(s)$ alone is the impulse response transform.
- (C) Wrong - dividing by s squared gives the ramp response.
- (D) Wrong - multiplying by s differentiates, giving the derivative of the impulse response.

A12. Correct answer: (A) $t \cdot u(t) - 2(t-1)u(t-1) + (t-2)u(t-2)$

- (A) Correct — the coefficient of each ramp is the change in slope, which is -2 at $t = 1$.
- (B) Wrong - this omits the final ramp and the waveform never returns to zero slope.
- (C) Wrong - using -1 instead of -2 makes the waveform flatten rather than descend.
- (D) Wrong - steps create jumps, not slope changes, so this produces a rectangular staircase.

A13. Correct answer: (A) multiplication

- (A) Correct — $Y(s) = H(s) \cdot X(s)$, which is why the transform route is so much cheaper.
- (B) Wrong - addition in s corresponds to addition in time, by linearity.
- (C) Wrong - the whole benefit of the transform is that convolution becomes multiplication.
- (D) Wrong - division in s corresponds to deconvolution, the inverse problem.

A14. Correct answer: (A) $s = -1000 \text{ rad/s}$

- (A) Correct — the pole sits at $-1/RC = -1000 \text{ rad/s}$.
- (B) Wrong - 159.2 is the cutoff expressed in hertz, not the pole location in rad/s .
- (C) Wrong - this is RC itself rather than its reciprocal.
- (D) Wrong - a right-half-plane pole would mean an unstable circuit, which a passive RC cannot be.

A15. Correct answer: (A) 159.2 Hz

- (A) Correct — $f_c = 1/(2\pi RC) = 1000/(2\pi) = 159.2 \text{ Hz}$.
- (B) Wrong - 1000 is the cutoff in rad/s ; the conversion to hertz was omitted.
- (C) Wrong - this multiplies by 2π instead of dividing.
- (D) Wrong - this is off by three orders of magnitude.

A16. Correct answer: (A) 0.35

- (A) Correct — this is why a fast circuit must also be a wideband circuit.

- (B) Wrong - 1.0 would follow only if bandwidth were expressed in rad/s and rise time as a time constant.
- (C) Wrong - 2.2 is the ratio of rise time to time constant, not this product.
- (D) Wrong - 6.28 is 2 pi and appears in the rad/s-to-hertz conversion, not here.

A17. Correct answer: (A) the complex pair at $-100 \pm j400$

- (A) Correct — the pole nearest the $j\omega$ axis decays slowest and therefore persists longest.
- (B) Wrong - a pole far from the axis decays fastest and contributes only a brief initial transient.
- (C) Wrong - largest magnitude means fastest decay, which is the opposite of dominance.
- (D) Wrong - the 20:1 ratio in decay rates makes one clearly dominant.

A18. Correct answer: (A) an undamped sustained oscillation

- (A) Correct — zero real part means zero decay, so the oscillation never dies.
- (B) Wrong - decay requires a strictly negative real part.
- (C) Wrong - growth requires a positive real part, which a passive network cannot produce.
- (D) Wrong - the pole contributes a very much present sustained oscillation.

A19. Correct answer: (A) not the square of the single-section $H(s)$, because of loading

- (A) Correct — the second section draws current from the first, shifting the pole locations.
- (B) Wrong - squaring is valid only when a buffer isolates the sections.
- (C) Wrong - cascading multiplies transfer functions; it never adds them.
- (D) Wrong - the second section clearly changes the response, just not in the naive way.

A20. Correct answer: (A) no right-half-plane poles or zeros and $\text{Re}\{Z\} \geq 0$ for $\text{Re}\{s\} \geq 0$

- (A) Correct — this is the realisability test for a passive one-port.
- (B) Wrong - poles on the axis correspond to lossless resonance and are a special case, not the general requirement.
- (C) Wrong - that condition makes the function improper and is not required.
- (D) Wrong - real coefficients are necessary but nowhere near sufficient.

Part B — model answers and marking scheme

B1 (3 marks) — CO4, L3, Ch.14, pp.553-561

Question. State the differentiation theorem and explain how it makes initial conditions automatic in circuit analysis.

Model answer. The theorem states that df/dt transforms to $sF(s) - f(0-)$. The $sF(s)$ term converts differentiation into multiplication, turning a differential equation into an algebraic one. The $-f(0-)$ term is the initial condition itself, deposited directly into the algebra by the transform. As a result a single s -domain solve yields the complete response, natural and forced together, with no separate boundary-condition stage — which is why the lower limit must be taken at $0-$.

Point required	Marks
Correct statement of the theorem	1
Explanation of the algebraic conversion and the role of $f(0-)$	2
Total	3

B2 (3 marks) — CO4, L4, Ch.14, pp.546-561

Question. Transform $2 \cdot di/dt + 6i = 12 \cdot u(t)$ with $i(0^-) = 3$ A, solve for $I(s)$, and find $i(t)$.

Model answer. Applying the differentiation theorem: $2[sI(s) - 3] + 6I(s) = 12/s$, so $I(s)(2s + 6) = 12/s + 6$ and $I(s) = (6s + 12)/(2s(s+3)) = (3s + 6)/(s(s+3))$. Partial fractions: $A = 6/3 = 2$ at $s = 0$ and $B = (-9+6)/(-3) = 1$ at $s = -3$, giving $I(s) = 2/s + 1/(s+3)$. Therefore $i(t) = 2 + e^{(-3t)}$ A, which starts at 3 A and settles at 2 A.

Point required	Marks
Correct transformed equation including the initial-condition term	1
Correct $I(s)$ and partial fraction residues	1
Correct $i(t)$ with both endpoints checked	1
Total	3

B3 (3 marks) — CO4, L4, Ch.14, pp.561-564

Question. State both value theorems, apply them to $F(s) = (3s + 6)/(s(s + 3))$, and state the condition that governs the final-value theorem.

Model answer. $f(0+) = \lim(s \rightarrow \infty) s \cdot F(s)$ and $f(\infty) = \lim(s \rightarrow 0) s \cdot F(s)$. Here $s \cdot F(s) = (3s + 6)/(s + 3)$. As $s \rightarrow \infty$ this tends to 3, so $f(0+) = 3$. The only pole of $s \cdot F(s)$ is at $s = -3$, strictly in the left half plane, so the final-value theorem applies; as $s \rightarrow 0$ the expression tends to 2, so $f(\infty) = 2$. If any pole of $s \cdot F(s)$ lay on the $j\omega$ axis or in the right half plane the limit would still return a number, but that number would be meaningless.

Point required	Marks
Both theorems stated correctly	1
Both limits evaluated correctly	1
Correct validity condition with the reason it matters	1
Total	3

B4 (3 marks) — CO4, L4, Ch.15, pp.588-602

Question. Derive the transfer function of a series RC low-pass with the output across the capacitor, and relate its pole to the time constant and the cutoff frequency.

Model answer. $H(s) = (1/sC)/(R + 1/sC) = 1/(1 + sRC) = (1/RC)/(s + 1/RC)$. There is a single pole at $s = -1/RC$ and no finite zeros. The pole distance from the origin is simultaneously $1/\tau$, since $\tau = RC$, and the -3 dB cutoff in rad/s, since $|H(j\omega)|$ falls to $1/\sqrt{2}$ at $\omega = 1/RC$. In hertz, $f_c = 1/(2\pi RC)$. For $R = 1$ k Ω and $C = 1$ μ F this gives a pole at -1000 rad/s, $\tau = 1$ ms and $f_c = 159.2$ Hz.

Point required	Marks
Correct derivation of $H(s)$	1
Pole identified and linked to τ	1
Cutoff computed in both rad/s and hertz	1
Total	3

B5 (2 marks) — CO4, L3, Ch.15, pp.571-578

Question. Explain how an inductor and a capacitor with non-zero initial conditions are modelled in the s -domain.

Model answer. An inductor becomes sL in series with a voltage source of value $L i(0^-)$, or equivalently sL in parallel with a current source $i(0^-)/s$. A capacitor becomes $1/(sC)$ in series with a voltage source $v(0^-)/s$, or $1/(sC)$ in parallel with a current source $C v(0^-)$. Once these models are inserted the network is a pure impedance network with sources, so one algebraic solve yields the complete response.

Point required	Marks
Both inductor forms correct	1
Both capacitor forms correct	1
Total	2

B6 (2 marks) — CO4, L3, Ch.14, pp.546-553

Question. Perform the partial fraction expansion of $(s + 4)/(s(s + 2))$ and write the time function.

Model answer. Write $F(s) = A/s + B/(s+2)$. Covering up s and evaluating the remainder at $s = 0$ gives $A = 4/2 = 2$. Covering up $(s+2)$ and evaluating at $s = -2$ gives $B = (-2+4)/(-2) = -1$. So $F(s) = 2/s - 1/(s+2)$ and $f(t) = 2 - e^{-2t}$ for $t \geq 0$. Checking with the initial-value theorem, $sF(s) \rightarrow 1$ as $s \rightarrow \infty$, and $f(0) = 2 - 1 = 1$, which agrees.

Point required	Marks
Both residues correct by the cover-up rule	1
Correct time function with a verification	1
Total	2

B7 (2 marks) — CO4, L3, Ch.14, pp.543-546

Question. Explain the relationship between the impulse, step and ramp responses, both in time and in the s -domain.

Model answer. The step is the integral of the impulse and the ramp is the integral of the step, so the same relationship holds between their responses: the step response is the integral of the impulse response, and the ramp response the integral of the step response. In the s -domain each integration divides by s , so the three transforms are $H(s)$, $H(s)/s$ and $H(s)/s^2$. Computing any one therefore determines the other two without further circuit analysis.

Point required	Marks
Correct integration relationship in the time domain	1
Correct corresponding division by s in the transform domain	1
Total	2

B8 (2 marks) — CO4, L4, TB2 §6.8

Question. Synthesise the transform of a triangular pulse that rises from 0 to 1 over $0 \leq t \leq 1$ and falls back to 0 by $t = 2$.

Model answer. Walk the waveform and add a ramp at each slope change, using the change in slope as the coefficient. Slope +1 begins at $t = 0$: add $t \cdot u(t)$. At $t = 1$ the slope goes from +1 to -1, a change of -2: add $-2(t-1)u(t-1)$. At $t = 2$ the slope goes from -1 to 0, a change of +1: add $(t-2)u(t-2)$. Applying the time-shift theorem, $F(s) = (1 - 2e^{-s} + e^{-2s})/s^2 = (1 - e^{-s})^2/s^2$.

Point required	Marks
Correct decomposition using changes in slope	1
Correct transform with the shift factors	1
Total	2

B9 (2 marks) — CO4, L5, Ch.15, pp.589-598

Question. Convolve $h(t) = e^{-t}u(t)$ with $x(t) = u(t)$, and verify the result in the s-domain.

Model answer. $y(t) = \int_0^t e^{-\tau} \cdot 1 \, d\tau = [-e^{-\tau}]_0^t = 1 - e^{-t}$. In the s-domain, $H(s) = 1/(s+1)$ and $X(s) = 1/s$, so $Y(s) = 1/(s(s+1)) = 1/s - 1/(s+1)$, which inverts to the same $1 - e^{-t}$. The s-domain route took one line against three, which is the usual reason to prefer it.

Point required	Marks
Correct convolution integral and result	1
Correct s-domain verification	1
Total	2

B10 (2 marks) — CO4, L5, Ch.15, pp.602-606

Question. Given poles at $s = -2000$ and $s = -100 \pm j400$, describe the natural response and identify the dominant pole.

Model answer. The real pole at -2000 gives an exponential with a 0.5 ms time constant — a brief initial transient. The complex pair gives a damped oscillation ringing at 400 rad/s inside an envelope e^{-100t} , a 10 ms time constant. Since the complex pair is nearest the $j\omega$ axis it decays slowest and therefore dominates the long-term response. Its damping ratio is $100/\sqrt{100^2 + 400^2} = 0.243$, so the ringing is lightly damped.

Point required	Marks
Correct description of both pole contributions	1
Correct identification of the dominant pole with the reason	1
Total	2

B11 (2 marks) — CO4, L5, Ch.15, pp.598-602

Question. Explain why the bandwidth-rise time product is constant for a first-order circuit, and state its value.

Model answer. The single pole sits at $-1/RC$, and that one number is both the cutoff in rad/s and the reciprocal of the time constant. The 10–90 % rise time is $2.2 \cdot RC$ and the cutoff in hertz is $1/(2\pi RC)$, so their product is $2.2/(2\pi) \approx 0.35$ regardless of the component values. Moving the pole further from the origin widens the bandwidth and shortens the rise time in exactly the same proportion, which is why a fast circuit must also be a wideband one.

Point required	Marks
Correct link between the single pole, bandwidth and rise time	1
Correct product value with its engineering consequence	1
Total	2

B12 (2 marks) — CO4, L5, Ch.15, pp.606-610

Question. Realise $H(s) = 1000/(s + 1000)$ as a passive RC network and explain why cascading two such sections does not give $H(s)^2$.

Model answer. Recognise the standard form $1/(1 + sRC)$ with $1/RC = 1000$. Choosing $C = 1 \mu\text{F}$ gives $R = 1 \text{ k}\Omega$, realised as a series resistor with the output across the capacitor. Forward analysis confirms $H(s) = 1000/(s + 1000)$. Cascading a second identical section does not square the response, because the second section draws current from the first and changes its effective load, shifting the pole locations. Squaring is valid only if a buffer isolates the two stages.

Point required	Marks
Correct component values with a forward verification	1
Correct loading explanation and the buffering remedy	1
Total	2

B13 (2 marks) — CO4, L5, Ch.15, pp.578-585

Question. Explain why the s-domain approach is preferred over the time-domain approach for a driven second-order circuit with non-zero initial conditions.

Model answer. In the time domain the natural response form must be selected from the damping classification, the forced response found separately, and two initial conditions applied to the combined expression — three distinct stages with a separate derivative evaluation. In the s-domain the initial conditions enter as sources, the impedances are algebraic, and a single solve yields a rational function whose partial fraction expansion contains the natural, forced and initial-condition contributions already combined. One pass replaces three, and the pole locations that had to be inferred from α and ω_0 appear directly as denominator roots.

Point required	Marks
Correct account of the time-domain stages	1
Correct account of why the s-domain collapses them into one	1
Total	2